FRAN IRISO

EMBEDDED SYSTEMS, ROBOTICS, MECHATRONICS, BIOMEDICAL ENGINEER

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SUMMARY

Engineer completing a dual Master's in Autonomous Systems and Intelligent Robots (July 2026), bringing a multidisciplinary background across robotics, electronics, and medical devices to tackle complex challenges with creativity. Internationally experienced, I combine strong technical execution with cross-cultural communication to deliver impact in diverse teams.

LANGUAGES

Spanish – Native | **Catalan –** Native | **English –** Full Professional Fluency
Italian – Intermediate | **Swedish –** Beginner | **French –** Beginner

PUBLICATIONS

"*Estimation Of Tidal Volume From Indirect Respiratory Temperature Signals Using Linear Regression*" **(Iriso, F. et al., 2026) – [*Pending publication* in IEEE EMBC 2026] (**Affiliation**:** Universitat Politècnica de Catalunya).

EXPERIENCE

Embedded Software Engineer Intern – Capsule Corporation srl, Milan, Italy February 2026 – July 2026
- Developed embedded software for a satellite water-propulsion system, focusing on real-time operation and reliability
- Designed and implemented FDIR (Fault Detection, Isolation & Recovery) in an RTOS-based modules strategy
- Conducted functional validation and debugging on avionics hardware, including stress and edge-case testing

Embedded Developer – Creekside AB, Stockholm, Sweden June 2023 – Sep 2024
- Designed and developed embedded software and hardware solutions across multiple industry applications
- Collaborated in cross-functional teams to deliver robust, production-ready solutions
- Contributed to projects for high-profile clients (e.g., *Epic Games*)

Research Engineer – Royal Institute of Technology (KTH), Stockholm, Sweden Jan 2023 – Aug 2023
- Developed a multi-dimensional in-ear sensing system as part of a Digital Futures research project
- Designed and implemented signal acquisition and processing methods for physiological data
- Collaborated with the principal investigator on medical device research in partnership with Karolinska Institutet (KI)
- Contributed to experimental validation and analysis of sensor performance in real-world conditions

Mathematics Teacher – Àrea d'Estudi, Barcelona, Spain July 2019 – Aug 2022
- Designed lesson plans in calculus, geometry, and algebra for students aged 15–23; 94% improved academic performance

EDUCATION

MSc: Mechatronics Engineering – University of Trento (UNITN, Italy) Sep 2024 - June 2026
GPA: 4.67/5.0 – EIT Digital Master School: Autonomous Systems and Intelligent Robots
Coursework: Distributed Robot Perception, Intelligent Vehicles, Path Planning

MSc: Computer Engineering – Budapest University of Technology and Economics (BME, Hungary) Sep 2024 - June 2026
GPA: 4.67/5.0 – EIT Digital Master School: Autonomous Systems and Intelligent Robots
Coursework: Machine Learning, AI Control, Computer Vision, Robotics

Minor of Innovation and Entrepreneurship – University of Rennes (France) June 2025 – July 2025
EIT Digital Master School

BSc: Biomedical Engineering – Universitat Politècnica de Catalunya (UPC, Barcelona) Sep 2021 – June 2025
GPA: 7.62/10 – 24 ECTS Thesis: *Multivariable model to estimate tidal volume from indirect respiratory signals.*

BSc: Industrial, Electronics Automatic and Control Engineering – UPC, Barcelona Sep 2018 – Jan 2024
GPA: 7.78/10 – 24 ECTS Thesis: *PCB Design for Remote Monitoring of a Motor-Controlled Polarizing Filter.*

Exchange Studies – Royal Institute of Technology (KTH), Stockholm June 2022 – Jan 2023

SKILLS

Programming: Python, C, C++, C#, MATLAB, SQL | **Embedded & Firmware:** STM32, ESP32, RTOS, Embedded Linux | **Robotics & Autonomy:** ROS2, Path Planning, Distributed Systems, Sensor Fusion, Perception Systems, Computer Vision | **Systems**: Control Systems, Signal Processing, FDIR, Real-Time Systems, State Machines, Docker | **Hardware & Prototyping:** PCB Design, Soldering, Circuit Design, Sensors and Actuators Integration, 3D CAD | **AI & Data:** Machine Learning, Data Analysis, Regression Models, Codex, Claude Code | **Other:** Medical Devices, Clinical Safety, Teamwork

ADDITIONAL INFORMATION

European Driver's License, U18 Spanish national Waterpolo Team. Spanish citizenship.